

Assam Floods and Flood Management in India



For Prelims: Southwest Monsoon , Ganga, Brahmaputra , National Perspective Plan , National Disaster Management Authority , Dam Safety Act, 2021 .

For Mains: Flood Management in India: Challenges and Way Forward, Climate Change and Increasing Flood Vulnerability, Disaster Management Framework in India

Source: TH

Why in News?

Assam has witnessed its worst floods in over 60 years , affecting **over 8 lakh people** and claiming **more than 30 lives** , highlighting persistent challenges in **India's flood management** and the need for **climate-resilient infrastructure** .

Summary

- Floods in India are increasingly driven by a combination of climate change, ecological degradation, unplanned urbanization, and governance gaps, exposing weaknesses in flood forecasting, reservoir management, interstate coordination, and disaster preparedness.
- Building flood resilience requires integrated river basin management, scientific reservoir operations, floodplain regulation, technology-driven early warning systems, nature-based solutions, and stronger institutional coordination to shift from disaster response to disaster risk reduction.

Why is Assam Highly Prone to Floods?

- **Geographical Vulnerability:** Assam lies in the **Brahmaputra and Barak river basins** , with over **50 tributaries** draining into them. Its location in the Brahmaputra floodplain makes it naturally prone to recurrent floods.
 - The **Brahmaputra carries one of the world's highest sediment loads** . As the river enters Assam's plains, reduced gradient causes sediment deposition, raising riverbeds and increasing flood frequency.

- **Heavy Monsoon Rainfall:** Intense southwest monsoon rainfall, coupled with heavy upstream discharge from **Arunachal Pradesh and Meghalaya** , frequently triggers flash floods.
- **Riverbank Erosion:** The dynamic nature of the Brahmaputra leads to frequent channel shifts, embankment breaches, and severe erosion. **About 4.27 lakh hectares** of land have been eroded since **1950** .
 - Deforestation and soil erosion in upstream catchments increase sediment inflow, reducing river carrying capacity and aggravating floods.
- **Climate Change:** Increasing extreme rainfall events and higher flood peaks due to climate change are intensifying the scale and frequency of floods.
- **Limited Effectiveness of Embankments:** Assam has relied heavily on embankments, many of which have exceeded their design life and are ineffective against the highly dynamic Brahmaputra.
- **High Flood Exposure:** **Nearly 39.58% of Assam's geographical area is flood-prone** , almost **four times the national average** , making floods an annual challenge.

What are the Causes of Floods in India?

- **Flood Vulnerability in India:** India has **over 40 million hectares** of flood-prone area out of its **329 million hectares** of geographical area.
 - On average, floods affect **75 lakh hectares annually** , claiming around **1,600 lives** , with major floods occurring **more than once every five years** .
 - Floods are also increasingly affecting regions previously considered non-flood-prone.



Natural Causes

- **Concentrated Monsoonal Rainfall:** Nearly **75-80% of India's annual rainfall** occurs during the **Southwest Monsoon (June-September)** .
 - High-intensity rainfall often exceeds the carrying capacity of rivers and drainage systems, resulting in widespread flooding.
- **Heavy Siltation and Riverbed Aggradation:** Young Himalayan rivers such as the **Ganga and Brahmaputra** carry enormous sediment loads.
 - Deposition of silt on riverbeds (**aggradation**) reduces channel capacity, increasing the frequency of overbank flooding.
- **River Meandering and Course Changes:** Highly sediment-laden rivers frequently alter their channels.
 - For example, the **Kosi River ("Sorrow of Bihar")** has repeatedly shifted its course, inundating new floodplains.
- **Glacial Lake Outburst Floods (GLOFs):** Accelerated glacier retreat due to **climate change** has expanded glacial lakes in the Himalayas, increasing the risk of sudden downstream floods, as witnessed during the **2023 Sikkim GLOF** .
- **Cyclones and Storm Surges:** Tropical cyclones along the **Bay of Bengal** generate intense rainfall and storm surges that inundate coastal regions while restricting river discharge into the sea.

Anthropogenic Causes

- **Deforestation and Catchment Degradation:** Large-scale deforestation in the **Himalayas and Western Ghats** increases **surface runoff**, accelerates soil erosion, and raises sediment deposition in rivers.
- **The 'Embankment Trap':** India has constructed **over 34,000 km of embankments** since 1954.
 - Excessive dependence on embankments often promotes **riverbed aggradation** , and breaches can result in sudden, prolonged, and more destructive floods.
- **Reservoir and Dam Mismanagement:** Inadequate adherence to **reservoir rule curves** and sudden releases during peak monsoon can aggravate downstream flooding, as observed during the **Kerala floods (2018)** .
- **Encroachment of Floodplains and Wetlands:** Unregulated construction on floodplains and degradation of **wetlands** such as **Assam's Beels** reduce the natural storage capacity of river systems, increasing flood intensity.
- **Unscientific Sand Mining:** Excessive sand extraction alters river morphology, destabilises riverbanks, and disrupts natural flow patterns, increasing flood vulnerability.

Urban Flooding

- **Unplanned Urbanisation:** Rapid concretisation has replaced permeable surfaces with asphalt and concrete, significantly increasing surface runoff.
- **Loss of Urban Water Bodies:** Encroachment of lakes, wetlands, and natural drainage channels has reduced cities' flood-buffering capacity, as seen in **Bengaluru** and **Chennai** .
- **Inadequate Storm-Water Drainage:** Ageing drainage infrastructure, combined with blockage by **solid waste and construction debris** , severely restricts storm-water evacuation during extreme rainfall events.
- **Climate-Induced Extreme Rainfall:** Increasing instances of short-duration, high-intensity rainfall are **overwhelming urban drainage systems, leading to frequent flash floods.**

What Measures has India Taken for Flood Management?

- **Structural Measures:**

- **Embankments and Drainage Works:** India has constructed **over 34,000 km of embankments** , drainage channels, flood protection works, and raised villages to reduce flood risks.
- **Dam Safety Act, 2021:** Establishes an institutional framework for the **surveillance, inspection, operation, and maintenance of specified dams** to enhance dam safety and reduce flood risks.
- **Reservoir Management:** Reservoirs are operated using Rule Curves (predefined target water levels) to regulate storage, maintain flood cushions, and moderate floods during the monsoon.
- **Inter-Basin Water Transfer (IBWT):** The **National Perspective Plan (1980)** envisages linking rivers to transfer surplus floodwaters to water-deficit regions.

- **Non-Structural Measures:**

- **Flood Plain Zoning:** The **Central Water Commission (CWC)** circulated the **Model Flood Plain Zoning Bill (1975)** to regulate development in flood-prone areas and minimise flood damage.
- **Flood Forecasting and Early Warning:** The **CWC** operates a nationwide flood forecasting network using **real-time telemetry, hydrological modelling, and inundation forecasting** .
- **Catchment Area Treatment (CAT):** Afforestation, watershed management, and soil conservation measures are undertaken to reduce runoff, soil erosion, and sedimentation.
- **Community-Based Disaster Preparedness:** Awareness programmes, mock drills, and local disaster management plans enhance community resilience.

- **Institutional and Policy Measures:**

- **Disaster Management Act, 2005:** Provides the legal framework for disaster preparedness, mitigation, response, and recovery through the **National Disaster Management Authority (NDMA)** , **State Disaster Management Authorities (SDMAs)** , **District Disaster Management Authorities (DDMAs)** , **National Disaster Response Force (NDRF)** , and **State Disaster Response Forces (SDRFs)** .
- **Flood Management and Border Areas Programme (FMBAP):** A **Ministry of Jal Shakti** scheme providing central assistance for **flood control, anti-erosion works, drainage development, and anti-sea erosion projects** .
- **National Water Policy, 2012:** Promotes **integrated river basin management** , flood moderation, watershed conservation, and restoration of natural drainage systems.
- **India Meteorological Department (IMD):** Provides weather forecasts and **impact-based warnings** to support early evacuation and disaster preparedness.

What are the Concerns Regarding India's Flood Management?

- **Constitutional and Federal Gridlock:** India's flood governance suffers from constitutional fragmentation.
 - **Entry 17 of the State List** gives states control over water, drainage and embankments, while **Entry 56 of the Union List** empowers the Centre over inter-state rivers.
 - This division leads to poor coordination, with upstream states often prioritizing irrigation and hydropower over downstream flood safety.

- Despite the **River Boards Act, 1956** , empowered River Basin Organizations have not been operationalized, preventing integrated basin-level flood management.
- **Fiscal and Project Execution Failures:** Performance Audits by the **Comptroller and Auditor General (CAG)** reveal serious weaknesses in the **Flood Management Programme (FMP)** .
 - Delays in approving **Detailed Project Reports (DPRs)** make project designs outdated before implementation, while the absence of basin-wide planning reduces effectiveness.
 - Delayed fund releases by the Centre, diversion of funds by states and poor utilization leave many flood-control and anti-erosion projects incomplete.
- **Deficient Flood Forecasting and Telemetry:** India's flood forecasting system is constrained by weak monitoring infrastructure.
 - Many telemetry stations and automated water-level sensors installed by the **CWC** remain non-functional because of poor maintenance, lack of calibration and vandalism.
 - Delayed or unreliable hydrological data limits accurate forecasting and reduces the lead time available for evacuation and reservoir management.
- **Transboundary Water Governance Gaps :** Flood management is further weakened by inadequate international cooperation. The **Indus Waters Treaty (1960)** focuses mainly on water sharing and hydropower, without addressing climate change, glacial melt or flood management.
 - In the Brahmaputra basin, India depends on **seasonal MoUs with China for hydrological data rather than a binding treaty** , limiting transparency over upstream water releases and weakening early-warning systems in the Northeast.
- **Weak Dam Safety and Reservoir Operations:** Poor adherence to reservoir operation protocols often converts natural floods into man-made disasters. Reservoir operators frequently violate **Rule Curves** by storing excess water to maximize irrigation and hydropower benefits.
 - During extreme rainfall, sudden large-scale releases from dams intensify downstream flooding. Moreover, many dams still lack legally enforced **Emergency Action Plans (EAPs)** and effective public warning mechanisms despite the **Dam Safety Act, 2021** .
- **Weak Grassroots Disaster Preparedness:** Disaster response at the local level remains inadequate due to limited institutional capacity.
 - Many states underfund and under-equip their **State Disaster Response Forces (SDRFs)** and district administrations, resulting in weak community preparedness, insufficient rescue equipment and poor early-warning dissemination.
 - Consequently, states remain heavily dependent on the **National Disaster Response Force (NDRF)** after disasters occur rather than investing in proactive local resilience.
- **Concerns Related to Northeast India:** The Northeast's flood management is constrained by its fragile **Himalayan ecology, transboundary river systems** .
 - Major rivers originate in **China, Bhutan** limiting India's control over upstream water flows and data availability.
 - Weak hydrological cooperation with neighbouring countries, inadequate forecasting infrastructure, difficult terrain, and poor connectivity further hinder integrated river basin management, disaster response, and long-term flood resilience, resulting in **recurring loss of lives, livelihoods, biodiversity, and critical infrastructure**.

What Measures Can Strengthen India's Flood Management?

- **Shift Towards Flood Resilience:** Adopt the recommendations of the **Rashtriya Barh Ayog (National Flood Commission), 1980** , which discourage indiscriminate embankments and promote floodplain management, raised settlements, flood-resilient agriculture, and coexistence with floods.
 - **R. Rangachari Committee (2003)** recommended a shift from **flood control to flood management**, emphasizing **Integrated River Basin Management (IRBM)**, floodplain zoning, catchment area treatment, reservoir regulation, improved forecasting, and community participation.
- **Improve Reservoir and Flood Forecasting:** Maintain a **dynamic flood cushion** in reservoirs by revising Rule Curves, develop a **National Water Model (NWM)** and **Decision Support System (DSS)** for real-time forecasting, and modernize hydro-meteorological networks, as recommended by the **NITI Aayog Committee on Flood Management (2021)** .
- **Strengthen Disaster Risk Reduction:** Shift from a relief-centric approach to **risk reduction** , investing in resilient infrastructure, preparedness, early warning systems, and the **Build Back Better** principle in line with the **Sendai Framework for Disaster Risk Reduction (2015-2030)**.
- **Regulate Floodplain Development:** Encourage states to implement **floodplain zoning laws** based on the **Model Flood Plain Zoning Bill (1975)** through financial incentives and strict regulation of construction on floodplains, wetlands, and natural drainage channels.
- **Promote Basin-Level Water Governance** : Establish **River Basin Organizations (RBOs)** to enable integrated river basin management through coordinated planning, reservoir operations, sediment management, and interstate data sharing.
- **Leverage Advanced Technology:** Expand **AI-based Early Warning Systems (EWS)** , IoT-enabled monitoring of embankments and reservoirs, satellite-based flood mapping, and strengthen the **CWC National Flood Forecasting Network**
- **Adopt Nature-Based Solutions** : Implement the 'Room for the River' approach and **Sponge City** initiatives by restoring wetlands, reconnecting floodplains, rejuvenating urban water bodies, and increasing permeable surfaces to reduce flood peaks.
- **Interlinking of Rivers (ILR):** It can reduce floods by storing excess monsoon water in **multi-purpose reservoirs** on rivers like the **Kosi** and **Mahanadi** , moderating downstream flood peaks through regulated releases.
 - However, it must be complemented by **floodplain zoning, embankments, and early warning systems** , as it cannot eliminate floods entirely.

Conclusion

Floods in India are no longer just an annual natural anomaly—they have transformed into recurring humanitarian and economic crises aggravated by climate change and unplanned urbanization. Mitigating this vulnerability requires breaking away from short-term embankments and relief payouts. A sustainable future demands an integrated, proactive approach centered on ecological restoration, strict floodplain governance, and advanced technological forecasting.

Drishti Mains Question:

Floods in India are increasingly becoming governance failures rather than purely natural disasters. Examine.

Frequently Asked Questions (FAQs)

1. What is the objective of the Flood Management and Border Areas Programme (FMBAP)?

It is a **Ministry of Jal Shakti** scheme that provides central assistance for flood control, anti-erosion measures, drainage development, and anti-sea erosion works.

2. What is the significance of the Dam Safety Act, 2021?

It establishes an institutional framework for the **surveillance, inspection, operation, and maintenance** of specified dams to improve dam safety and reduce flood risks.

3. What is Flood Plain Zoning?

Flood Plain Zoning regulates development in flood-prone areas to minimize flood damage and prevent encroachment on floodplains and wetlands. It is based on the **Model Flood Plain Zoning Bill (1975)**.

4. Why is Integrated River Basin Management (IRBM) important?

IRBM promotes coordinated planning, reservoir operations, sediment management, and data sharing across entire river basins, reducing interstate conflicts and improving flood resilience.

[Watch Video on YouTube:

▶ https://www.youtube.com/embed/dBYM_wyGglg

]

UPSC Civil Services Examination, Previous Years Questions (PYQs)

Prelims:

Q. La Nina is suspected to have caused recent floods in Australia. How is La Nina different from El Nino? (2011)

1. La Nina is characterised by an usually cold ocean temperature in equatorial Indian Ocean whereas El Nino is characterised by unusually warm ocean temperature in the equatorial Pacific Ocean.
2. El Nino has adverse effect on south-west monsoon of India but La Nina has no effect on monsoon climate.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

Ans: (d)

Mains :

Q. The interlinking of rivers can provide viable solutions to the multi-dimensional inter-related problems of droughts, floods, and interrupted navigation. Critically examine. (2020)

Q. Account for the huge flooding of million cities in India including the smart ones like Hyderabad and Pune. Suggest lasting remedial measures. (2020)

Early Release Policy for Elderly and Terminally Ill Prisoners



Source: TH

Why in News?

The **Supreme Court** directed all States and Union Territories to **frame and notify, within three months, a uniform policy** for the premature release of elderly and terminally ill prisoners on humanitarian grounds.

- The order was passed while hearing a petition filed by the **National Legal Services Authority (NALSA)**, which highlighted the absence of a common framework for such prisoners.
- NALSA argued that facilitating the release of such inmates would address **humanitarian concerns** and help reduce prison overcrowding.

What Did the Supreme Court Direct for Premature Release of Elderly and Terminally Ill Prisoners?

- **Objective:** The directions aim to ensure **humane treatment of vulnerable prisoners, reduce prison overcrowding**, improve access to justice, and establish a uniform national mechanism for compassionate release.
- **Eligibility Criteria and Procedure:** Every State policy must clearly define eligibility conditions, application procedures, and timelines for considering remission or premature release requests.
 - The policies must be framed in consultation with the respective **State Legal Services Authorities (SLSAs)** to ensure coordination and effective identification of eligible prisoners.
- **Medical Assessment Mechanism:** States and UTs must constitute **independent medical boards** to assess and certify cases of terminal illness, advanced age, and severe medical vulnerability while considering applications for remission.
- **Definition of Terminal Illness:** The Court directed that policies must include a uniform definition of “terminal illness” and allowed authorities to adopt the definition provided in the **UN Office on Drugs and Crime (UNODC) Handbook on Prisoners with Special Needs**, which refers to conditions with no reasonable medical possibility of preventing continued deterioration leading to death.
- **Role of Under Trial Review Committees (UTRCs):** UTRCs must periodically review cases of prisoners who are terminally ill, elderly, or suffering from physical incapacitation and recommend appropriate legal measures such as **bail, parole, remission, or release**.
- **Time-Bound Decision-Making:** The Court directed authorities to avoid procedural delays that may

result in prisoners spending their final months or years in incarceration. All decisions must be reasoned and remain subject to judicial review.

- **Digital Tracking System:** The Union Government, in coordination with the **National Informatics Centre (NIC)** , must provide technical assistance, digital infrastructure, and training to States and UTs and operationalise an **e-portal** for processing and tracking premature release applications.
- **Transparency and Accountability:** The e-portal must enable automated alerts, monitoring of prescribed timelines, generation of compliance reports, and supervisory oversight while ensuring confidentiality and privacy of prisoners' medical and personal information.
- **Compliance Monitoring:** The Centre, States, and UTs must file compliance affidavits within **six months** , providing details of prisoners identified under the policy, those granted premature release, and cases still under consideration.

Status of Prison Overcrowding in India

- At the end of 2024, India operated **1,333 jails with a collective sanctioned capacity of 4.53 lakh inmates** .
 - However, the actual **inmate population exceeded 5.11 lakh** , forcing the system to run well over its intended limits.
- More than **half of the States and Union Territories (UTs) registered occupancy rates exceeding 100%** in 2024.
 - **Delhi recorded the highest prison occupancy rate in India** at 194.6% in 2024, followed by Meghalaya (163.5%), Jammu & Kashmir (148.3%), and Madhya Pradesh (147.1%).
 - **Jammu and Kashmir** also witnessed a massive surge, with occupancy rates compounding from a mere 78% in 2015 to **over 148%** in 2023 and 2024.
 - Conversely, states like **Chhattisgarh drastically reduced overcrowding** , dropping its occupancy rate from an extreme 234% in 2015 to **127.6%** in 2024, alongside similar downward trends in Uttar Pradesh.

What Steps Have the Government Taken to Address Prison Overcrowding?

- **Bharatiya Nagarik Suraksha Sanhita (BNSS), 2023:**
 - **Section 479(1):** A progressive step replacing **CrPC** 436A. It states that **first-time offenders** who have served **one-third** of their maximum prescribed sentence must be released on bond.
 - It also mandates that the jail superintendent must officially apply for the undertrial's release once they reach the stipulated time limit.
- **Model Prisons and Correctional Services Act, 2023:** Replaces the archaic **Prisons Act of 1894**.
 - It shifts the focus from **"retribution" to "reformation and rehabilitation"** and introduces concepts like electronic tagging for prisoners on parole/furlough.
- **Prison Reforms:** The **2016 Model Prison Manual** was introduced to standardise prison management, focusing on prisoner classification, medical care, and vocational training.
 - In 2018, the **Prisons Development Fund** was launched to modernize prison infrastructure and

support state-level improvements.

- **E-Prisons Project:** Digitization of prison records to seamlessly integrate with the **Inter-operable Criminal Justice System (ICJS)**, ensuring timely alerts when an inmate becomes eligible for bail.
- **Support to Poor Prisoners Scheme:** Launched by the Ministry of Home Affairs to provide financial assistance to marginalized undertrials who are unable to pay their bail amounts or secure sureties.

National Legal Services Authority (NALSA)

- **About:** NALSA is a **statutory body** constituted in **1995** under the **Legal Services Authorities Act, 1987**, to **monitor, evaluate, and implement** legal aid programmes across India. It has completed **30 years** in 2025.
 - The **Chief Justice of India (CJI)** is the **Patron-in-Chief** of NALSA.
- **Constitutional Mandate:** Formed to give effect to **Article 39A** of the Constitution of India, which mandates the State to provide **free legal aid** to ensure **equal justice and opportunities** for all citizens, particularly those with economic or other disabilities.
 - Also upholds the obligations under **Article 14** (equality before law) and **Article 22(1)** (rights to be informed of grounds for arrest).
- **Primary Functions:** Lay down **policies and principles** for making **legal services available** under the Legal Services Authorities Act, 1987.
 - **Monitor and evaluate** the implementation of legal aid programmes across India.
 - **Disburse funds and grants** to State Legal Services Authorities (SLSAs) and **Non-Governmental Organisations (NGOs)** for implementing legal aid schemes and programmes.
- **Integrated Network:** NALSA is the **apex body** of a nationwide network envisaged under the Legal Services Authorities Act, 1987 for providing **legal aid and assistance**. This network includes:
 - **State Legal Services Authorities (SLSAs)** : Headed by the **Chief Justice** of the respective **High Court** (Patron-in-Chief).
 - **District Legal Services Authorities (DLSAs)** : Chaired by the **District Judge** of the respective district.
 - **Taluk/Sub-Divisional Legal Services Committees** : Headed by a **senior civil judge**.
 - High Court Legal Services Committees and the Supreme Court Legal Services Committee.
- **Eligible Groups:** Women and Children, Members of **Scheduled Castes (SC)** and **Scheduled Tribes (ST)**, **Economically Weaker Sections (EWS)**, Industrial Workers, Disabled Persons and Other specified categories.

Conclusion

The Supreme Court's direction marks an important step towards humane prison administration and rights-based criminal justice reform. By mandating State-level policies, medical assessment, legal aid coordination and digital tracking, the Court has sought to ensure dignity, compassion and accountability for elderly and terminally ill prisoners. Effective implementation will be crucial to transform this judgment into meaningful relief.

Drishti Mains Question

The humane treatment of prisoners is an essential part of constitutional governance. Discuss in the context of the Supreme Court's directions on early release of elderly and terminally ill prisoners

Frequently Asked Questions (FAQs)

1. What did the Supreme Court direct?

It directed all States and Union Territories to frame a policy within three months for early or premature release of elderly and terminally ill prisoners.

2. Which body filed the petition?

The petition was filed by the National Legal Services Authority.

3. What role will SLSAs play?

State Legal Services Authorities will help frame and implement policies and identify eligible prisoners.

4. What is the role of medical boards?

Independent medical boards will assess and certify terminal illness or serious medical vulnerability.

5. Why is the e-Prisons portal important?

It will enable time-bound processing, automated alerts, transparency, compliance reports and institutional oversight.

[Watch Video on YouTube:

▶ <https://www.youtube.com/embed/j9FB0YCbVjg>

]

UPSC Civil Services Examination, Previous Year Questions

Prelims

Q. Consider the following statements:

Statement-I: In India, prisons are managed by State Governments with their own rules and regulations for the day-to-day administration of prisons.

Statement-II: In India, prisons are governed by the Prisons Act, 1894 which expressly kept the subject of prisons in the control of Provincial Governments.

Which one of the following is correct in respect of the above statements?

- a) Both Statement-I and Statement-II are correct and Statement-II is the correct explanation for Statement-I
- b) Both Statement-I and Statement-II are correct and Statement-II is not the correct explanation for Statement-I
- c) Statement-I is correct but Statement-II is incorrect
- d) Statement-I is incorrect but Statement-II is correct

Ans: (a)

Q. With reference to India, consider the following statements: (2021)

1. When a prisoner makes out a sufficient case, parole cannot be denied to such prisoner because it becomes a matter of his/her right.
2. State Governments have their own Prisoners Release on Parole Rules.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Ans: (b)

Mains

Q. National Human Rights Commission (NHRC) in India can be most effective when its tasks are adequately supported by other mechanisms that ensure the accountability of a government. In light of above observation assess the role of NHRC as an effective complement to the judiciary and other institutions in promoting and protecting human rights standards. **(2014)**



Source: IE

Why in News?

India's **Commonwealth Games 2026 campaign** suffered a setback even before the opening ceremony as two medal prospects were removed due to anti-doping violations, affecting the country's medal expectations.

- This comes in the background of the Government of India notifying the commencement of the **National Anti-Doping Act, 2022** and the **National Anti-Doping (Amendment) Act, 2025**, aimed at strengthening India's legal and institutional framework for clean sport.

Why has India's Doping Issue Gained Attention?

- **About:** India's weightlifting quota was **reduced from 16 to 11 athletes** due to multiple anti-doping violations within the specified period. Under the **Commonwealth Games qualification rules**, every anti-doping violation by an athlete or support personnel leads to a reduction in the number of athlete slots allocated to that country.
- India topped the list of doping offenders in the **2024 annual report of the World Anti-Doping Agency (WADA)**, with a **positivity ratio of 3.6%**, among the highest globally.
- The issue assumes greater importance as **India prepares to host the 2030 Commonwealth Games** and continues its bid for the 2036 Olympics, highlighting the need for stronger anti-doping mechanisms.

Note: Doping refers to the use of prohibited substances or methods by athletes to enhance performance.

What are the Key Features of the National Anti-Doping (Amendment) Act, 2025?

- **About:** The **National Anti-Doping (Amendment) Act, 2025** amends the **National Anti-Doping Act, 2022** to strengthen India's anti-doping framework and promote clean sport.
 - It provides a legal framework to prevent doping, protect athletes' rights, and maintain the integrity of sporting competitions.
 - The Act aligns India's anti-doping system with the **World Anti-Doping Code** and the **UNESCO International Convention against Doping in Sport**.
- **Institutional Framework:**
 - Establishes the **National Anti-Doping Agency (NADA)** for implementing anti-doping rules.
 - Establishes the **National Board for Anti-Doping in Sports** to oversee NADA and advise the Central Government on anti-doping regulations.
- **Key Provisions:**
 - Provides statutory backing for testing, results management, education, intelligence, and investigations related to doping violations.
 - Empowers the National Board to obtain information from the Disciplinary Panel and Appeal Panel

regarding their operations.

- Defines conduct and circumstances that constitute anti-doping rule violations.

- **Significance:**

- Strengthens transparency, accountability, and due process in anti-doping administration.
- Enhances institutional autonomy and brings India's anti-doping framework closer to international best practices.
- Promotes fair play and ensures a level playing field for Indian athletes.

National Anti-Doping Agency (NADA)

- **About :** NADA is **India's autonomous body** , established in **2005** , responsible for promoting and monitoring **anti-doping** in **sports** .
- **Functions:** NADA is responsible for planning, implementing, and coordinating India's anti-doping activities. It **adheres to the World Anti-Doping Agency (WADA) code and regulations**.
- **Initiatives:** It implements the **Anti-Doping Program** and conducts initiatives like the **#PlayTrue Campaign** to educate athletes and promote **clean competition**.

World Anti-Doping Agency (WADA)

- **About: WADA** , founded in **1999** by the **International Olympic Committee (IOC)** , is the **global independent authority** tasked with **promoting and coordinating the fight against doping in sports**.
- **Governance Model:** Operates on a unique principle of **equal partnership** and **funding** between the **global Sport Movement** and **Governments** .
- **Core Document:** The **World Anti-Doping Code** is its **fundamental charter** , designed to **harmonize anti-doping rules** across all **sports** and **nations** for **consistent athlete standards** .
- **Key Standard:** It publishes the **WADA Prohibited List** , the **international standard** identifying **banned substances** and **methods** , which is **updated annually** .
- **Operational Scope:** The **Prohibited List** and **Code** apply to **all scenarios** — **in-competition, out-of-competition** , and for **specific sports** .

Frequently Asked Questions (FAQs)

1. Why was India's Commonwealth Games 2026 campaign affected by doping violations?

India faced a setback after two medal prospects were removed due to anti-doping rule violations. Additionally, India's weightlifting quota was reduced from **16 to 11 athletes** due to multiple doping violations within the specified qualification period.

2. Why was India's weightlifting quota reduced?

Under the Commonwealth Games qualification rules, every **anti-doping rule violation (ADRV)** by an athlete or athlete support personnel results in a reduction of athlete slots allocated to that country.

3. Which Indian athlete was suspended before the Commonwealth Games 2026?

Arun Kumar, a judoka competing in the **78 kg category**, was provisionally suspended by the **National Anti-Doping Agency (NADA)**.

4. What is the National Anti-Doping Act, 2022?

The National Anti-Doping Act, 2022 provides a statutory framework for regulating anti-doping activities in India, including testing, investigations, result management and athlete education.

5. What changes were introduced through the National Anti-Doping (Amendment) Act, 2025?

The amendment strengthens institutional autonomy, improves procedural fairness, enhances governance mechanisms and aligns India's anti-doping system with evolving international standards.

[Watch Video on YouTube:

▶ <https://www.youtube.com/embed/ogWGZiSASHU>

]

UPSC Civil Services Examination, Previous Year Question (PYQ)

Prelims

Q. Consider the following statements in respect of the 32nd Summer Olympics. (2021)

1. The official motto for this Olympics is 'A New World'.
2. Sport Climbing, Surfing, Skateboarding, Karate and Baseball are included in this Olympics.

Which of the above statements is/are correct?

- (a) Only 1
- (b) Only 2
- (c) Both 1 and 2
- (d) Neither 1 nor 2

Ans. (b)

Mains

Q . An athlete participates in the Olympics for personal triumph and nation's glory; victors are showered with cash incentives by various agencies, on their return. Discuss the merit of state sponsored talent hunt and its cultivation as against the rationale of a reward mechanism as encouragement. **(2014)**

Space Oncology and Cancer Treatment



Source: TH

Why in News?

Cancer is a major global health challenge, with India expected to witness around 1.87 million new cases in 2026, creating significant health and economic burdens.

- In this context, **space oncology has emerged as a new frontier in cancer research**, using microgravity and space-based environments to study tumour behaviour, accelerate drug discovery, and develop advanced cancer therapies.

What is Space Oncology?

- **About:** Space oncology is an **emerging field that studies how space conditions, especially microgravity and cosmic radiation**, affect cancer cells, tumour growth, treatment response, and cancer-cell adaptation.
- **Space as a Natural Laboratory:** Space provides a unique environment to study cancer biology, helping researchers understand metastasis, treatment resistance, radiation response, biomarker discovery, and tissue behaviour.
- **Research Value:** It supports advanced **tumour modelling, drug discovery, protein crystallisation, drug formulation, 3D cell cultures**, and **tumour spheroid** development.

Importance of Space Oncology

- **Rising Cancer Burden:** Cancer involves uncontrolled growth of abnormal cells, and India is expected to record around **1.87 million new cases in 2026**.
- **Economic Burden:** Cancer imposes financial stress through **treatment costs, income loss, debt, asset depletion, and caregiver burden**.
- **Need for Advanced Models:** Space-based environments provide **improved tumour models** that can reveal cancer mechanisms and accelerate therapy development.
- **Drug Development Potential:** Microgravity can improve **drug formulation, enhance testing models, and reduce dependence on animal studies**.

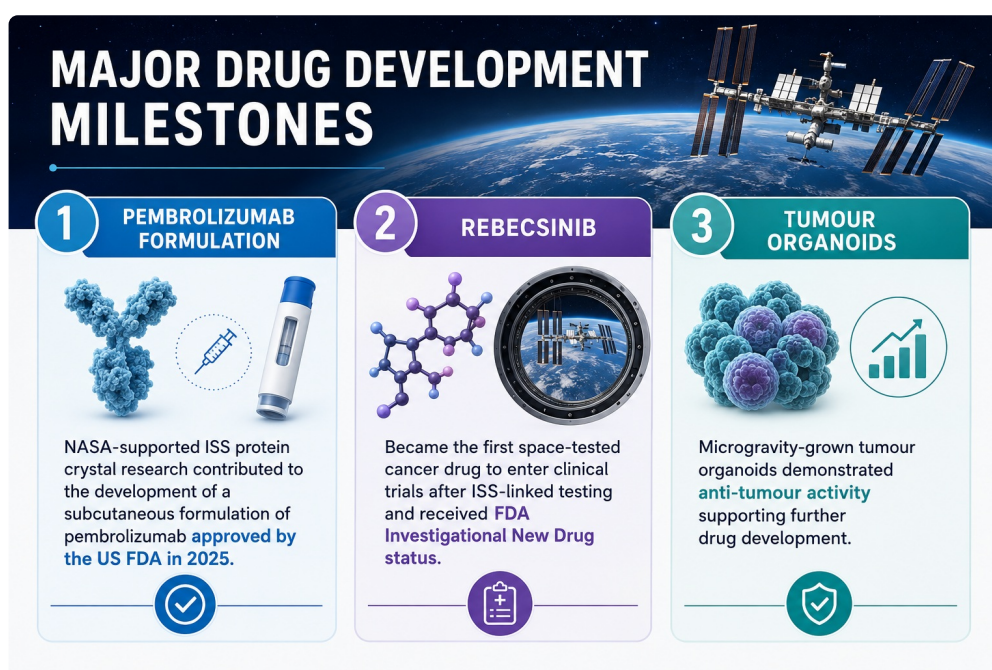
Microgravity Effects on Cancer Cells

- **Cellular Changes:** Microgravity causes **cytoskeletal reorganisation, altered focal adhesion signalling, and changes in extracellular matrix interaction**.

- **Tumour Behaviour:** It affects cancer cell adhesion, migration, proliferation, invasion, and gene expression.
- **Spheroid Formation:** Microgravity enables formation of multicellular tumour spheroids, improving cancer modelling.
- **Drug Response:** Changes in cell structure and gene expression influence drug uptake and treatment response.

Space Oncology and Cancer Types

- **Breast Cancer:** Microgravity can reduce malignant characteristics by affecting adhesion, migration, and cell-cycle regulation, making tumour spheroids more responsive to specialised therapies.
- **Gastrointestinal Cancer:** Some gastrointestinal cancers show altered progression under microgravity, with increased drug sensitivity due to changes in resistance genes and cellular proteins.



What Role can India Play in the Emerging Field of Space Oncology?

- **Leverage Space Capabilities:** India can utilise **ISRO's space expertise**, growing space economy, and emerging private space sector to develop space-based biomedical research capabilities.
- **Promote Space-Based Pharmaceutical Research:** India can encourage research on microgravity-based drug development, cancer models, and space manufacturing of advanced medicines.
- **Strengthen Public-Private Partnerships:** Collaboration between ISRO, research institutions, hospitals, pharmaceutical companies, and space startups can accelerate innovation in space oncology.
- **Develop Affordable Cancer Solutions:** India can focus on translating space-based research into cost-effective cancer diagnostics, therapies, and drug delivery systems to address the growing cancer burden.
- **Build Global Leadership:** With falling launch costs and increasing commercial space opportunities, India can emerge as a hub for space biotechnology and cancer research.

Frequently Asked Questions

1. What is space oncology?

Space oncology is the study of how microgravity and cosmic radiation affect cancer progression, tumour behaviour and treatment response.

2. Why is space useful for cancer research?

Space provides microgravity and radiation conditions that alter cancer-cell behaviour, helping scientists understand tumour biology in ways that are difficult on Earth.

3. How does microgravity affect cancer cells?

Microgravity can alter cytoskeleton structure, cell adhesion, gene expression, migration, proliferation and spheroid formation.

4. How can space oncology help drug discovery?

It can improve tumour modelling, protein crystal growth, nanoparticle drug delivery and testing of cancer therapies.

5. What are tumour spheroids?

Tumour spheroids are three-dimensional clusters of cancer cells that can better represent tumour-like behaviour compared to flat cell cultures.

[Watch Video on YouTube: [▶ https://www.youtube.com/embed/Dg-LiMr1fxk](https://www.youtube.com/embed/Dg-LiMr1fxk)]

UPSC Civil Services Examination, Previous Year Questions

Prelims

Q. With reference to India's satellite launch vehicles, consider the following statements: (2018)

1. PSLVs launch satellites useful for Earth resources monitoring whereas GSLVs are designed mainly to launch communication satellites.
2. Satellites launched by PSLV appear to remain permanently fixed in the same position in the sky, as viewed from a particular location on Earth.
3. GSLV Mk III is a four-staged launch vehicle with the first and third stages using solid rocket motors and the second and fourth stages using liquid rocket engines.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 and 3
- (c) 1 and 2
- (d) 3 only

Ans: (a)

Mains

Q. What is the basic principle behind vaccine development? How do vaccines work? What approaches were adopted by Indian vaccine manufacturers to produce COVID-19 vaccines? (2022)

23rd Commonwealth Games 2026



Source: AIR

The **23rd edition of the Commonwealth Games** is being held in **Glasgow, Scotland** over **10 days** (23rd July -2nd August 2026), with participation from teams representing **74 nations and territories** .

- **Featured Sports:** The 2026 edition features only **10 sports** , nine fewer than Birmingham 2022. The events include **athletics, swimming, 3×3 basketball, track cycling, weightlifting, bowls, artistic gymnastics, netball, boxing and judo** .
- **Indian Participation:** India is participating with a **125-member contingent, including para-athletes**.
 - Indian athletes will compete in athletics, boxing, weightlifting, judo, artistic gymnastics, track cycling, bowls, swimming, and para-sports. **India will not send teams for 3×3 basketball and netball.**
- **India's Participation and Performance:** India first participated in the Commonwealth Games in **1934** . Its best-ever performance came at the **2010 Commonwealth Games in Delhi** , where it won **101 medals** .

Commonwealth Games

- **About:** CWG is the **world's second-largest multi-sport event** after the Olympics, featuring athletes from 71 nations and territories, most of which are **former British colonies**.
 - CWG was the brainchild of Canadian sportswriter **Melville Marks Robinson** .
 - The idea of CWG originated from the 1911 Inter-Empire Championships in London, held during the Festival of Empire for King George V's coronation.
 - The first official CWG, then known as the **British Empire Games**, were **held in 1930** with **Canada's Hamilton as hosts** .
 - Canadian athlete **Gordon Smallacombe won the first gold medal** at the Commonwealth Games in the triple jump.
 - Since the first edition, the Games have been held **every four years** except in 1942 and 1946, and they were formally renamed the **Commonwealth Games in 1978**.
- **Governing Body:** **Commonwealth Games Federation (CGF)** is responsible for the direction and control of the Games.
- **India at CWG:** India last hosted the **Commonwealth Games in 2010 in New Delhi**, and has won the bid to host the **Centenary Commonwealth Games 2030, with Ahmedabad, Gujarat** selected as the host city.

Read More: [India to Host Commonwealth Games 2030](#)

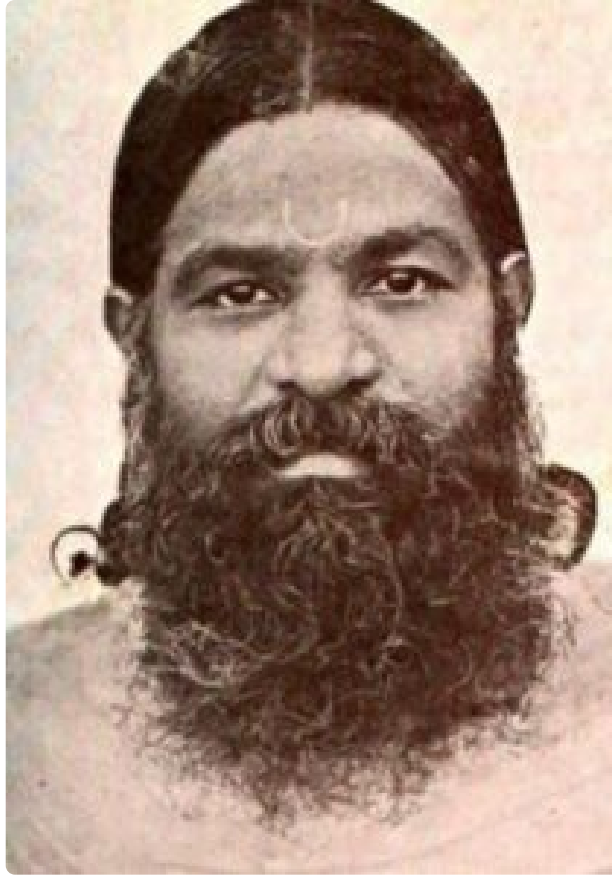
Death Anniversary of Thiru Subramania Siva



Source: PIB

Vice-President of India Shri C. P. Radhakrishnan paid floral tributes to the freedom fighter **Thiru Subramania Siva** on the occasion of his death anniversary (23 rd July).

- Subramania Siva was born on 4 th October 1884 at Batlagundu (present-day Dindigul district, Tamil Nadu).
- **Literary & Cultural Contributions:** Siva was a pioneer of the **Tanittamil (Pure Tamil) Movement** , advocating the use of Tamil free from excessive Sanskrit influence.
 - He became a disciple of Sadananda Swami in 1902, after which he adopted the name “Sivam” and developed a strong spiritual-nationalist outlook.
 - In 1913, he founded the nationalist journal **Gnanabhanu** . Across his life, he authored over 30 books, including **Ramanuja Vijayam, Madhva Vijayam, and Vitutalaipporil Tamil Valarnta Varalaru**
- **Nationalist Awakening:** Inspired by the nationalist speeches of **Arya Samaj leader Thakur Kahan Chandra Verma** , Siva became a powerful public speaker.
 - Along with **V.O. Chidambaram Pillai** and **Subramania Bharati** , he promoted **Bal Gangadhar Tilak's** nationalist ideology.
 - He also founded the **Dharma Paripalana Samajam (1906)** to promote social reform and patriotic awakening.
- **Role in the Freedom Movement:** Siva played a pivotal role in the **Swadeshi Movement** , working closely with **V.O. Chidambaram Pillai** . He mobilised workers during the historic **Coral Mill Strike (1908)** at **Thoothukudi** and organised public meetings despite British prohibitions.
 - His speeches transformed **Tirunelveli** into a major centre of nationalist activity, leading to his arrest for **sedition** , making him the **first recognised political prisoner of the Madras Presidency** .
 - In **1923** , Siva laid the foundation for the **Bharat Mata Temple** at **Pappalapatti (Dharmapuri)** , reflecting his vision of cultural nationalism.
 - **Subramania Siva** contracted **leprosy** due to harsh jail conditions. Despite ill health, he remained committed to the freedom struggle until his **death on 23rd July 1925** .



Read more: [Swadeshi and Boycott Movement](#)